ONLINE MOVIE TICKET RESERVATION SYSTEM

AI23431- WEBTECHNOLOGY AND MOBILE APPLICATION

PROJECT REPORT

SUBMITTED BY

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

RAJALAKSHMI ENGINEERING COLLEGE (AUTONOMOUS),

CHENNAI-602 105

APRIL 2026

RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project is titled "**ONLINE MOVIE TICKET RESERVATION**" is the Bonafide work of MUKHESH L (2116241501121),MUKESH RAGHAVENDHAR(241501120),PREMRAJ R(241501152) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.,** and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN**, **Ph.D.,** for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.,** our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. K. SEKAR, Ph.D.**, Professor and Head of the Department of Artificial Intelligence and Machine Learning for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, **Dr.A RAJASEKRAN,** Department of Artificial Intelligence and Machine Learning for his useful tips during our review to build our project.

MUKHESH L -241501121

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DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Machine Learning Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

COURSE OBJECTIVE

- To identify and formulate real-world problems that can be solved using Artificial Intelligence and Machine Learning techniques.
- To apply theoretical and practical knowledge of AI/ML for designing innovative, data-driven solutions.
- To integrate various tools, frameworks, and algorithms to develop, test, and validate AI/ML models.
- To demonstrate effective teamwork, project management, and communication skills through collaborative project execution.
- To instill awareness of ethical, societal, and environmental considerations in the design and deployment of intelligent systems.

COURSE OUTCOME

- **CO 1:** Analyze and define a real-world problem by identifying key challenges, project requirements and constraints.
- **CO 2:** Conduct a thorough literature review to evaluate existing solutions, identify research gaps and formulate research questions.
- **CO 3:** Develop a detailed project plan by defining objectives, setting timelines, and identifying key deliverables to guide the implementation process.
- **CO 4:** Design and implement a prototype or initial model based on the proposed solution framework using appropriate AI tools and technologies.

CO 5 : Demonstrate teamwork, communication, and project management skills by preparing and presenting a well-structured project proposal and initial implementation results

CO-PO-PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO 1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO 2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO 3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO 4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO 5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low)   2: Moderate (Medium)   3:
Substantial (High) No correlation: “-”

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

The **AI/ML Department Web Application** is a web-based system designed to streamline and manage academic activities within a department efficiently. The application provides a centralized platform for students, faculty, and administrators to access and manage essential academic information such as attendance, results, announcements, and placement details.

The system is developed using **Flask (Python)** for backend processing, along with **HTML, CSS, and JavaScript** for the frontend interface. It incorporates a structured database to store and retrieve data securely, ensuring reliability and scalability. The application features role-based access control, allowing different functionalities for students, faculty, and administrators.

Key functionalities include student attendance tracking, result management, faculty dashboards, and administrative controls for managing users and academic records. The system improves transparency, reduces manual effort, and enhances communication within the department.

Overall, the project demonstrates an effective implementation of a web-based solution for academic management, integrating modern web technologies to deliver a user-friendly and efficient system.

SI NO	TITLE	PAGE NO

LIST OF ABBREVIATIONS

SAMS – Student Attendance Management System

DB – Database

SQL – Structured Query Language

API – Application Programming Interface

UI – User Interface

HTML – HyperText Markup Language

JS – JavaScript

HTTP – HyperText Transfer Protocol

SMTP – Simple Mail Transfer Protocol

CRUD – Create, Read, Update, Delete

CHAPTER 1

INTRODUCTION

1. General

The **Online Movie Ticket Reservation System** is a web-based application designed to simplify and automate the process of booking movie tickets. In traditional booking methods, customers often face long queues, limited seat availability, and time constraints. This system eliminates such issues by providing an efficient and user-friendly digital platform where users can browse movies, select theatres, choose show timings, and reserve seats from anywhere at any time.

The application is developed using modern web technologies, with a responsive frontend interface for user interaction and a backend server to handle business logic and database operations. The system stores data such as user details, movie information, and booking records using a structured database, ensuring accuracy and easy retrieval of information. It also provides secure authentication for users and an admin panel for managing movies and monitoring bookings.

Additionally, the system enhances the user experience through features like seat selection, real-time availability, booking summaries, and ticket generation. By integrating these functionalities, the Online Movie Ticket Reservation System improves efficiency, reduces manual effort, and provides a convenient solution for both customers and administrators in managing movie ticket bookings effectively.

1.2 Need for AI/ML Department Web Application

The **Online Movie Ticket Reservation System** is highly relevant for the AIML (Artificial Intelligence and Machine Learning) department as it provides a practical platform to apply core concepts such as data handling, user behavior analysis, and system automation. Students can enhance the system by integrating AI features like movie recommendation systems, demand prediction, and dynamic pricing based on user preferences. It also helps in understanding real-world application development, combining frontend, backend, and database management. Such projects bridge theoretical AIML knowledge with practical implementation, improving problem-solving skills and preparing students for industry-level intelligent system development.

1.3 Scope of the System

The scope of the Online Movie Ticket Reservation System is broad and can be extended to include multiple advanced features and real-world applications. The system can be enhanced to support multiple cities, theatres, and a large number of users simultaneously, making it scalable for commercial use. It can be integrated with secure online payment gateways to enable real-time ticket booking and transactions.

From an AIML perspective, the system can be expanded by incorporating intelligent features such as personalized movie recommendations based on user preferences, predictive analysis for seat demand, and dynamic pricing strategies. It can also include analytics dashboards for administrators to monitor booking trends, revenue, and user behavior.

Additionally, the system can be adapted for mobile applications, real-time notifications, and integration with external APIs for movie details and reviews. Overall, the project has strong potential for future development into a full-fledged, industry-level application with smart automation and data-driven decision-making capabilities.

1.4 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the Online movie ticket reservation Web Application.

Chapter 1: provides the introduction to the project, including the general overview, need for the system, and scope of the project.

Chapter 2: presents the literature survey, which discusses existing attendance management systems and related work.

Chapter 3: explains the problem formulation and objectives of the proposed system.

Chapter 4: describes the system design, including architecture and system requirements.

Chapter 5: discusses the implementation of the system and explains the different modules.

Chapter 6 : presents the results and performance evaluation of the system.

Chapter 7 : concludes the project and discusses possible future enhancements.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The literature survey is an essential part of any project as it provides a comprehensive understanding of existing systems, technologies, and research related to the Online Movie Ticket Reservation System. It involves studying previously developed applications, methodologies, and tools used in similar domains to identify their strengths and limitations. Through this analysis, developers can gain insights into efficient system design, user interface development, database management, and security practices.

In the context of online ticket booking systems, many platforms have demonstrated the importance of real-time data processing, user authentication, and seamless interaction between frontend and backend components. The literature survey also highlights emerging trends such as automation, cloud-based systems, and integration of intelligent features. By reviewing these studies, the proposed system can be designed to overcome existing drawbacks and incorporate improved functionalities. Thus, the literature survey acts as a strong foundation for building a reliable, efficient, and user-friendly application.

2.2 Existing Department Management Systems

The existing system for managing movie ticket reservations is mostly manual or partially digital, often handled through counters or basic management software. In such systems, users must visit theatres physically to book tickets, leading to time consumption and inconvenience. Seat availability is not always updated in real time, and there is a higher chance of booking conflicts and errors. Additionally, record maintenance is done manually or with limited automation, making data retrieval and management difficult. These limitations highlight the need for a fully automated online reservation system that improves accessibility, accuracy, and efficiency.

2.3 Related Work

Several research studies and projects have been carried out in the field of online movie ticket booking systems, focusing on improving efficiency, usability, and system performance. Earlier systems were developed as web-based applications that allowed users to register, log in, browse movies, and book tickets through an online portal. These systems typically included two main modules: a user interface for customers and an admin panel for managing movies and bookings.

A comparative study on online movie booking systems highlights that with the growth of digitalization and internet usage, such systems have become highly popular, eliminating the need to stand in queues and enabling real-time seat selection. However, challenges such as server overload, transaction failures, and system scalability were identified, leading to the adoption of modern technologies like Node.js and MongoDB for improved performance.

2.4 Summary

From the literature survey, it is evident that modern web-based systems offer significant advantages over traditional methods. They improve accuracy, reduce manual workload, and enable centralized data management. However, many existing systems lack simplicity, flexibility, or affordability for specific departmental use.

Existing online movie ticket booking systems improve convenience by enabling users to book tickets, select seats, and access real-time information. However, challenges like scalability, security, and system performance still exist. These limitations highlight the need for enhanced systems with better efficiency, user experience, and intelligent features. Additionally, integration of AI and real-time analytics can further improve system accuracy and personalization.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

The traditional movie ticket booking process is often manual or partially digital, which leads to several inefficiencies and inconveniences for users. Customers are required to stand in long queues at theatres, and there is no proper way to check real-time seat availability before reaching the venue. This results in wasted time and uncertainty in booking preferred seats or show timings. Additionally, the absence of a centralized system makes it difficult to manage multiple theatres, movies, and schedules efficiently.

Furthermore, manual record-keeping increases the chances of errors, data inconsistency, and booking conflicts. Existing systems may lack proper user authentication, scalability, and an intuitive interface for smooth interaction. These limitations highlight the need for a fully automated and user-friendly Online Movie Ticket Reservation System that ensures real-time updates, secure transactions, accurate data management, and an improved overall booking experience for both users and administrators.

3.2 Objectives of the System

The main objective of the Online movie reservation system Web Application is to develop a digital platform that simplifies departmental management and improves accessibility to academic information.

- To develop a user-friendly web application that allows users to browse movies, select seats, and book tickets easily with real-time availability.
- To implement secure user authentication and efficient database management for storing user, movie, and booking details accurately.
- To provide an admin module for managing movies, monitoring bookings, and ensuring smooth system operation with improved efficiency and automation.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design defines the overall structure and working of the Online Movie Ticket Reservation System. It explains how different components such as the user interface, backend server, and database interact to perform various operations like user authentication, movie display, seat selection, and booking management. The design ensures smooth data flow, efficient processing, and proper integration of all modules. It also focuses on scalability, security, and usability to provide a reliable and user-friendly system that meets the functional and performance requirements.

4.2 System Architecture

The Online Movie Ticket Reservation System follows a three-tier architecture consisting of presentation, application, and data layers. The presentation layer includes the user interface developed using HTML, CSS, and JavaScript, allowing users to interact with the system.

The application layer is built using Flask, which processes user requests, handles business logic, and communicates between the frontend and database. The data layer uses SQLite to store user details, movie information, and booking records securely. This architecture ensures proper separation of concerns, improves scalability, and enhances system performance, making the application efficient, maintainable, and suitable for real-time ticket booking

4.3 System Requirements

A computer with a web browser, Python (Flask), and SQLite installed is required to run and manage the Online Movie Ticket Reservation.

The Online Movie reservation Web Application does not require high-end hardware and can run on standard computing systems.

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 GB Hard Disk or SSD
- Input Devices: Keyboard and Mouse
- Internet Connection

4.3.2 Software Requirements

The system is developed using commonly available software tools and technologies.

- Operating System: Windows 10 or later
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technology: Flask
- Database: SQLite / MySQL
- Development Environment: Visual Studio Code / Antigravity
- Web Browser: Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The implementation of the Online Movie Ticket Reservation System follows a structured and modular approach to ensure efficiency and reliability. Initially, the system requirements are analyzed, and the overall architecture is designed based on a three-tier model consisting of frontend, backend, and database layers. The frontend is developed using HTML, CSS, and JavaScript to create an interactive user interface for login, movie browsing, and seat selection.

The backend is implemented using Python with the Flask framework, which handles API requests, user authentication, and booking logic. SQLite is used as the database to store user details, movie information, and booking records. The system integrates these components through RESTful APIs to ensure smooth communication and real-time updates.

Testing is performed at each stage to verify functionality, identify errors, and improve performance. Finally, the system is deployed locally, ensuring proper execution of all modules and providing a seamless ticket booking experience.

5.2 Module Descriptions

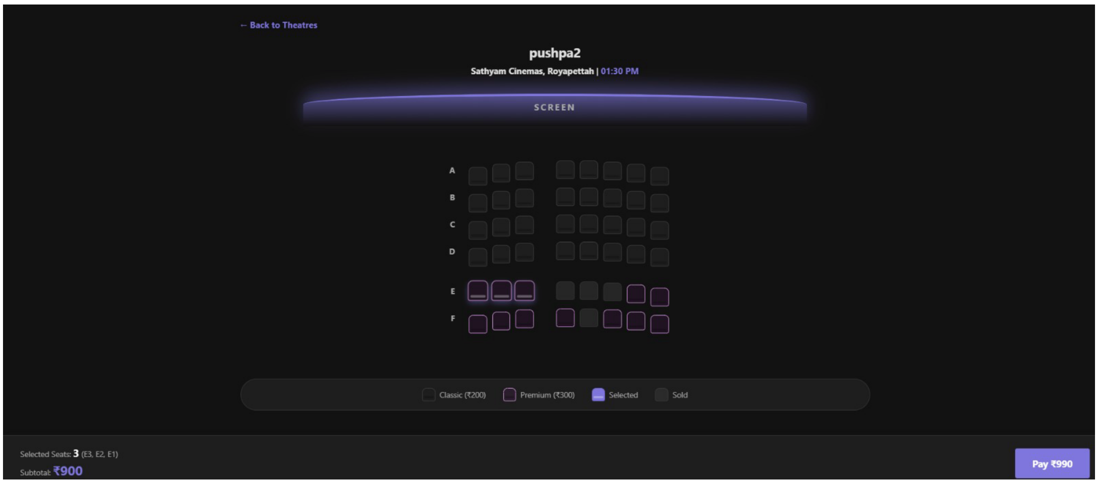
The Online movie reservation Web Application consists of several modules that work together to provide efficient departmental management.

Index Module

The Index Module serves as the main interface of the Online Movie Ticket Reservation System and is developed using HTML, CSS, and JavaScript. It acts as the entry point for users, providing options for login, registration, and admin access.

This module displays available movies, allows users to search based on name or genre, and supports city selection to view relevant theatres and show timings. It dynamically updates content through JavaScript by interacting with the backend APIs, ensuring real-time data display.

The module also manages user interactions such as seat selection, booking summary, and ticket generation. Overall, it provides a responsive, user-friendly environment and ensures smooth communication between the frontend and backend for efficient system operation. interface.



5.2.1 Movie Selection Module

The Movie Selection Module is a key component of the Online Movie Ticket Reservation System that allows users to browse and choose movies based on their preferences. This module displays a list of available movies along with essential details such as title, genre, duration, and ratings. Users can search for movies using keywords or filter them based on categories, improving ease of navigation. Once a movie is selected, the system retrieves and displays the corresponding theatres and available show timings based on the selected city.

The module dynamically updates movie information using backend API calls, ensuring real-time data accuracy. It provides an intuitive and visually appealing interface, making it easy for users to explore different options. Additionally, the module ensures smooth transition to the next stage of booking, where users can select theatres and seats. Overall, it enhances user experience by simplifying the movie selection process.

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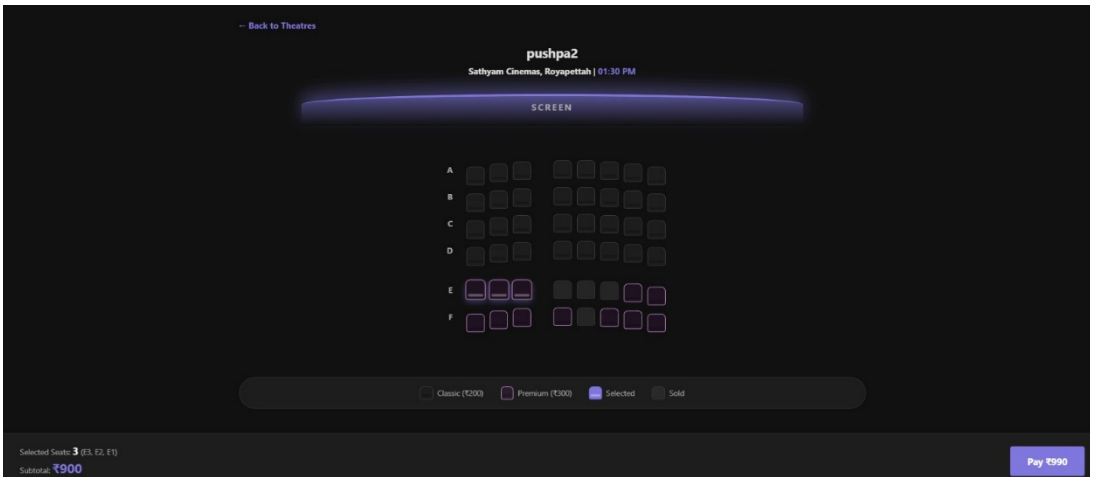
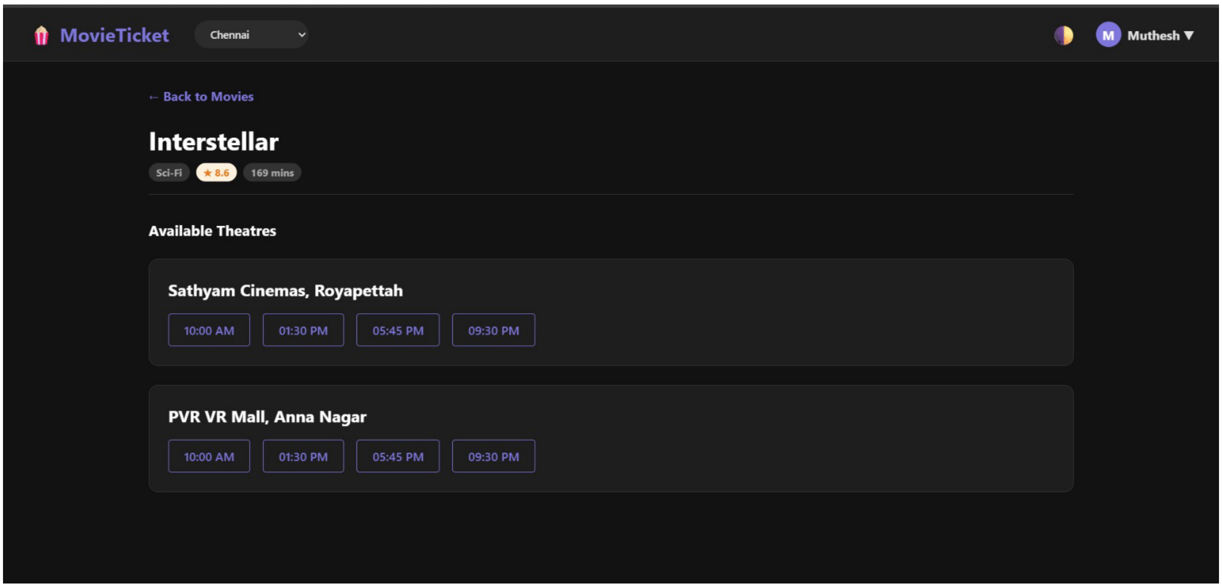


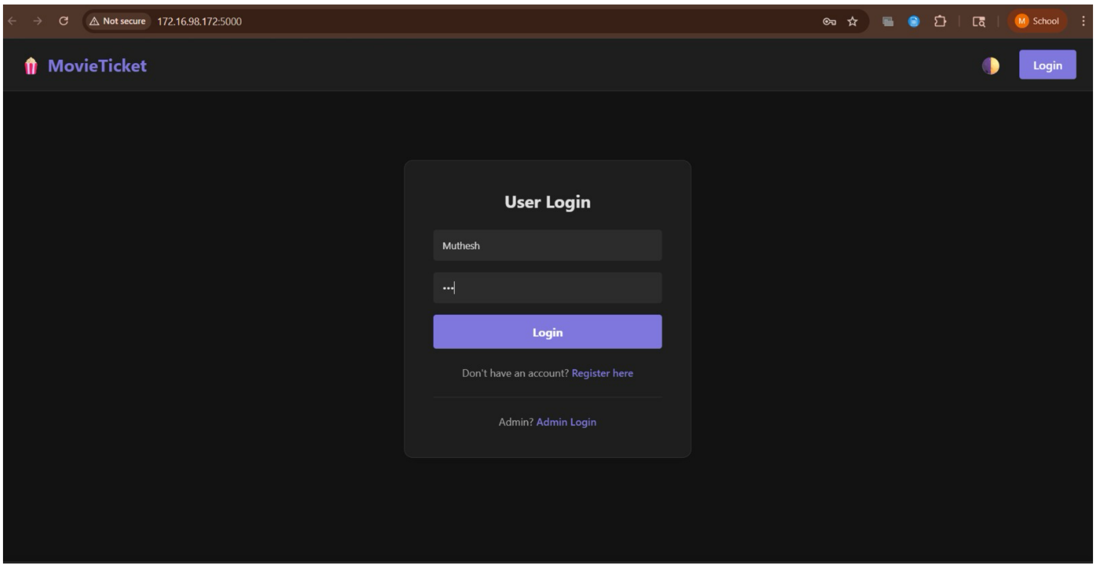
5.2.2 Theatre & Show Management Module

The This module displays theatres and available show timings based on the selected movie and city. It helps users choose convenient locations and timings. The system ensures real-time updates of show schedules and integrates seamlessly with the booking module for a smooth user experience



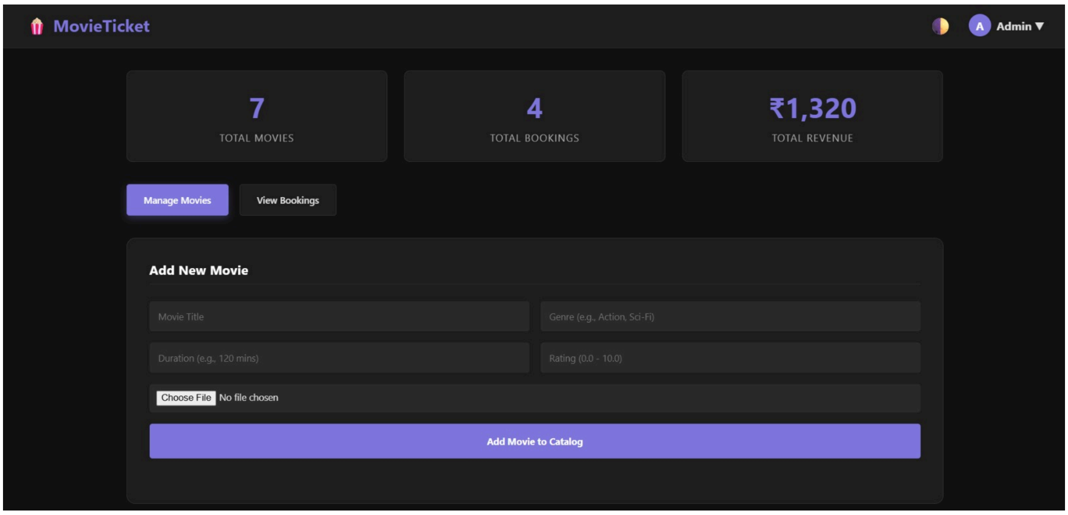
5.2.3 Login and Authentication Module

The Login Module allows registered users and admin to securely access the system using their username and password. It verifies credentials with the database and grants appropriate access based on user roles. This module ensures authentication, prevents unauthorized access, and maintains session management for a secure and personalized user experience.



5.2.4 Admin Module

This module allows administrators to manage the system effectively. Admin can add, update, or remove movies, and view all bookings. It provides insights into system performance, user activity, and revenue. This module ensures smooth operation, maintenance, and control of the entire booking system.



CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The Online Movie Ticket Reservation System was successfully developed and tested, achieving all the intended objectives. The system allows users to register, log in, browse movies, select theatres and show timings, and book tickets with real-time seat availability. The frontend interface is responsive and user-friendly, ensuring smooth navigation and interaction. The backend, implemented using Flask, efficiently handles user requests and database operations. SQLite database ensures accurate storage and retrieval of user, movie, and booking data. The admin module enables effective management of movies and monitoring of bookings. Overall, the system performs reliably with accurate results, minimal errors, and improved efficiency compared to manual booking methods.

6.2 Performance Evaluation

The performance of the Online Movie Ticket Reservation System was evaluated based on response time, accuracy, usability, and reliability. The system demonstrates fast response in loading movie details, processing user requests, and updating seat availability in real time. Database operations using SQLite are efficient, ensuring quick storage and retrieval of booking information. The system maintains high accuracy by preventing duplicate bookings and ensuring data consistency. Usability testing shows that the interface is user-friendly and easy to navigate for both users and administrators. The backend efficiently handles multiple requests without significant delay. Overall, the system performs effectively, providing a reliable and smooth ticket booking experience.

6.3 Resource Utilization

The Online Movie Ticket Reservation System utilizes system resources efficiently to ensure smooth performance. The application requires minimal hardware resources, as it is lightweight and runs on a local server using Flask and SQLite. CPU and memory usage remain low during normal operations such as user login, movie browsing, and ticket booking. The SQLite database efficiently handles data storage and retrieval without requiring heavy processing power. Network usage is also optimized, as only necessary data is exchanged between the frontend and backend through API calls. Overall, the system demonstrates effective utilization of resources, making it suitable for small to medium-scale applications.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The **Online Movie Ticket Reservation System** was successfully designed and implemented to provide a simple, efficient, and user-friendly platform for booking movie tickets. The system eliminates the drawbacks of traditional booking methods by enabling users to access real-time information about movies, theatres, show timings, and seat availability. It ensures secure user authentication, accurate data management, and smooth interaction between the frontend, backend, and database components.

The application effectively integrates all modules such as user login, movie selection, seat booking, and admin management, resulting in a reliable and functional system. It improves efficiency, reduces manual effort, and enhances user experience. Overall, the project meets its objectives and demonstrates the practical implementation of web technologies in solving real-world problems.

7.2 Future Work

The Online Movie Ticket Reservation System can be further enhanced by integrating advanced features and technologies. Future improvements may include the addition of secure online payment gateways for real-time transactions and mobile application support for better accessibility. The system can also incorporate AI-based recommendation systems to suggest movies based on user preferences and viewing history.

Additionally, features like push notifications for show updates, cancellation and refund management, and integration with external APIs for movie reviews and ratings can be implemented.

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[7] Research papers on Online Ticket Booking Systems – available on platforms like IEEE Xplore and ResearchGate.

APPENDICES

Appendix-1: source code pseudocode

System Overview

The system is a web-based application used to manage online movie ticket reservations. It allows users to register, log in, browse movies, select theatres and show timings, and book tickets. Admin can manage movies and view bookings. The system ensures secure access, real-time seat availability, and accurate data handling.

..

System Initialization Logic

START

Connect to SQLite database

IF database connection is successful

  Start Flask server on port 5000

  Load system homepage (index.html)

ELSE

  Display database connection error

END IF

END

User Login Logic

FUNCTION Login(username, password)

Check database for matching username and

password

```
IF user exists
   IF role = admin
      Redirect to Admin Dashboard
   ELSE
      Redirect to User Dashboard
   END IF
ELSE
   Display "Invalid Credentials"
END IF
END FUNCTION
```

Faculty Attendance Marking Logic

```
FUNCTION Register(username, password)

Check if username already exists

IF exists
   Display "User already exists"
ELSE
   Store user details in database
   Display "Registration Successful"
END IF
```

END FUNCTION

SEAT BOOKING LOGIC:

```
Display seat layout

User selects available seats

IF seats are available
   Calculate total price
   Proceed to booking summary
ELSE
   Display "Seats not available"
END IF

END
```

Movie selection logic

Display list of available movies

User selects a movie

Fetch theatres and show timings based on selected city

Display theatres and timings

END

Booking Confirmation Logic:

FUNCTION ConfirmBooking(details)

Store booking details in database

Update seat status

Generate booking ID

Display ticket confirmation

END FUNCTION

CONCLUTION:

The Online Movie Ticket Reservation System has been successfully designed and implemented to provide an efficient and user-friendly platform for booking movie tickets. The system overcomes the limitations of traditional booking methods by offering real-time access to movie details, theatre information, and seat availability. It ensures secure user authentication and accurate data management using a structured database.

The integration of frontend technologies with a Flask-based backend enables smooth communication and fast processing of user requests. Key features such as movie selection, seat booking, and ticket generation enhance the overall user experience. The admin module further supports effective management of movies and bookings.

Overall, the system meets its objectives and demonstrates a practical and scalable solution for modern ticket reservation needs.